

**NAME: MAYANK MEHTA**  
**USN: 1BF24CS171**  
**VACUUM CLEANER:**

CODE:

```
class VacuumEnvironment:
    def __init__(self):
        # Environment state: 'Clean' (0) or 'Dirty' (1)
        self.status = {'A': 'Dirty', 'B': 'Dirty'}
        self.agent_location = 'A'

    def is_dirty(self, location):
        return self.status[location] == 'Dirty'

    def clean(self, location):
        self.status[location] = 'Clean'

class ReflexVacuumAgent:
    def act(self, location, status):
        if status == 'Dirty':
            return 'Suck'
        elif location == 'A':
            return 'Right'
        elif location == 'B':
            return 'Left'

def run_simulation():
    env = VacuumEnvironment()
    agent = ReflexVacuumAgent()

    print(f"Initial Environment State: {env.status}")
    print(f"Agent starts at Location: {env.agent_location}\n")

    steps = 4
    for step in range(1, steps + 1):
        loc = env.agent_location
        status = env.status[loc]
        action = agent.act(loc, status)
```

```
    print(f"[Step {step}] Location: {loc} | Status: {status} | Action: {action}")

    if action == 'Suck':
        env.clean(loc)
    elif action == 'Right':
        env.agent_location = 'B'
    elif action == 'Left':
        env.agent_location = 'A'

    print(f"\nFinal Environment State: {env.status}")

run_simulation()
```

OUTPUT:

```
... Initial Environment State: {'A': 'Dirty', 'B': 'Dirty'}
Agent starts at Location: A

[Step 1] Location: A | Status: Dirty | Action: Suck
[Step 2] Location: A | Status: Clean | Action: Right
[Step 3] Location: B | Status: Dirty | Action: Suck
[Step 4] Location: B | Status: Clean | Action: Left

Final Environment State: {'A': 'Clean', 'B': 'Clean'}
```